SET PROJECTS

NAVRACHANA UNIVERSITY

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POWER ELECTRONICS STUDIO (STU601)

SINGLE-PHASE HIGH FREQUENCY INVERTER

Name of course : B-Tech EEE, Supervisor Name : Ketan Bhavsar, Ankit Chouhan

Student Name/Enrolment No. : Arafat Babar/23000159, Vedant Acharya/23000229, Ketan Pitroda/23000796, Pritha Patel/23001118, Raj Gediya/23001115, Krisha Pandya/23000386 , Yash Patel/23001111 , Dev Bhavsar 23001109

INTRODUCTION

Conventional square-wave and modified sine-wave inverters generate high harmonic distortion, which can affect the performance of sensitive electrical and electronic devices. To overcome this issue, this project presents the design and implementation of a 1 kW pure sine wave inverter using SPWM (Sinusoidal Pulse Width Modulation) and STM32G4 microcontroller-based control. The system uses a two-stage AC-DC-AC conversion process to produce stable, efficient, and low-distortion AC output with reliable switching control and safe isolated operation.

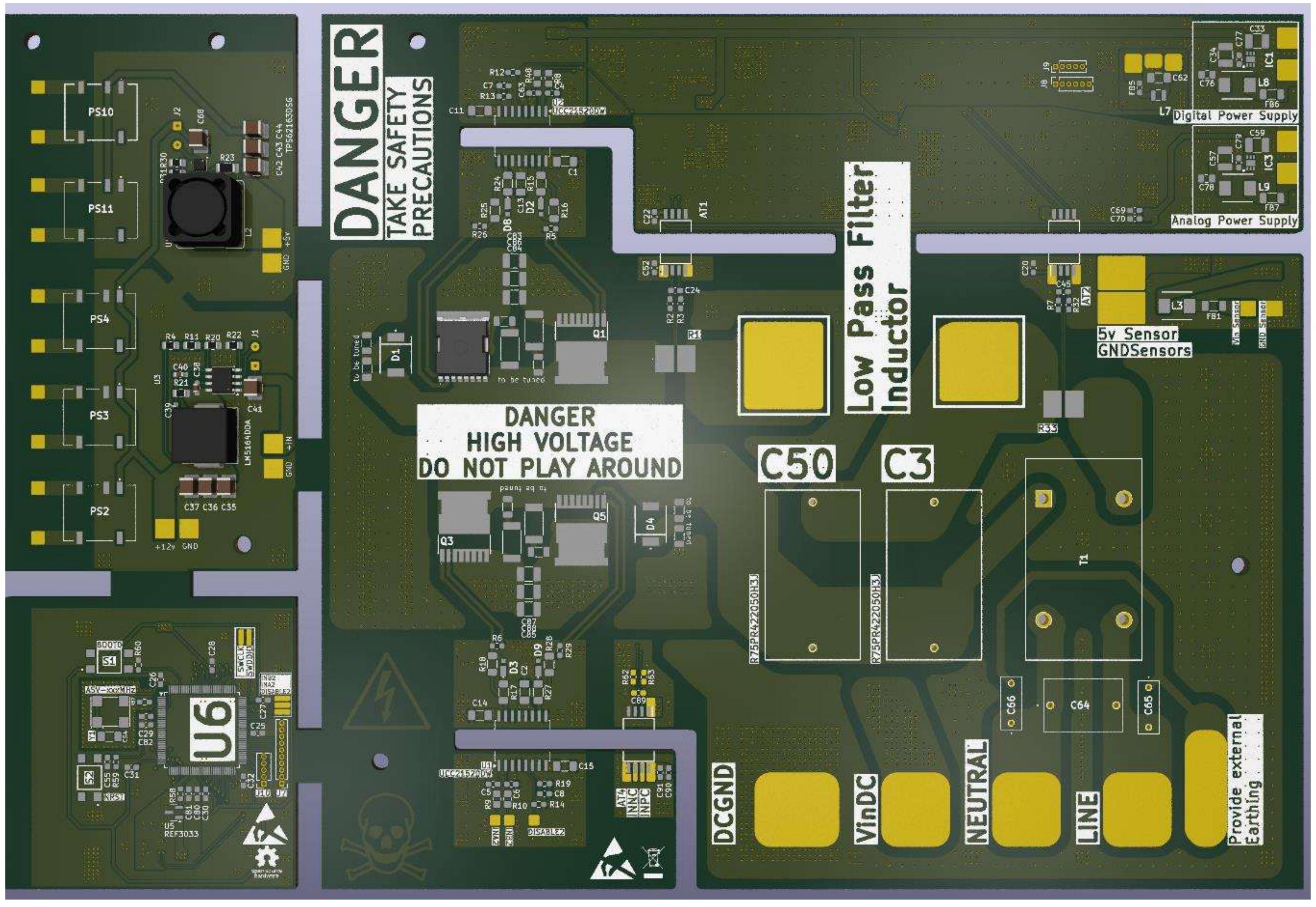

Fig .1 PCB Design

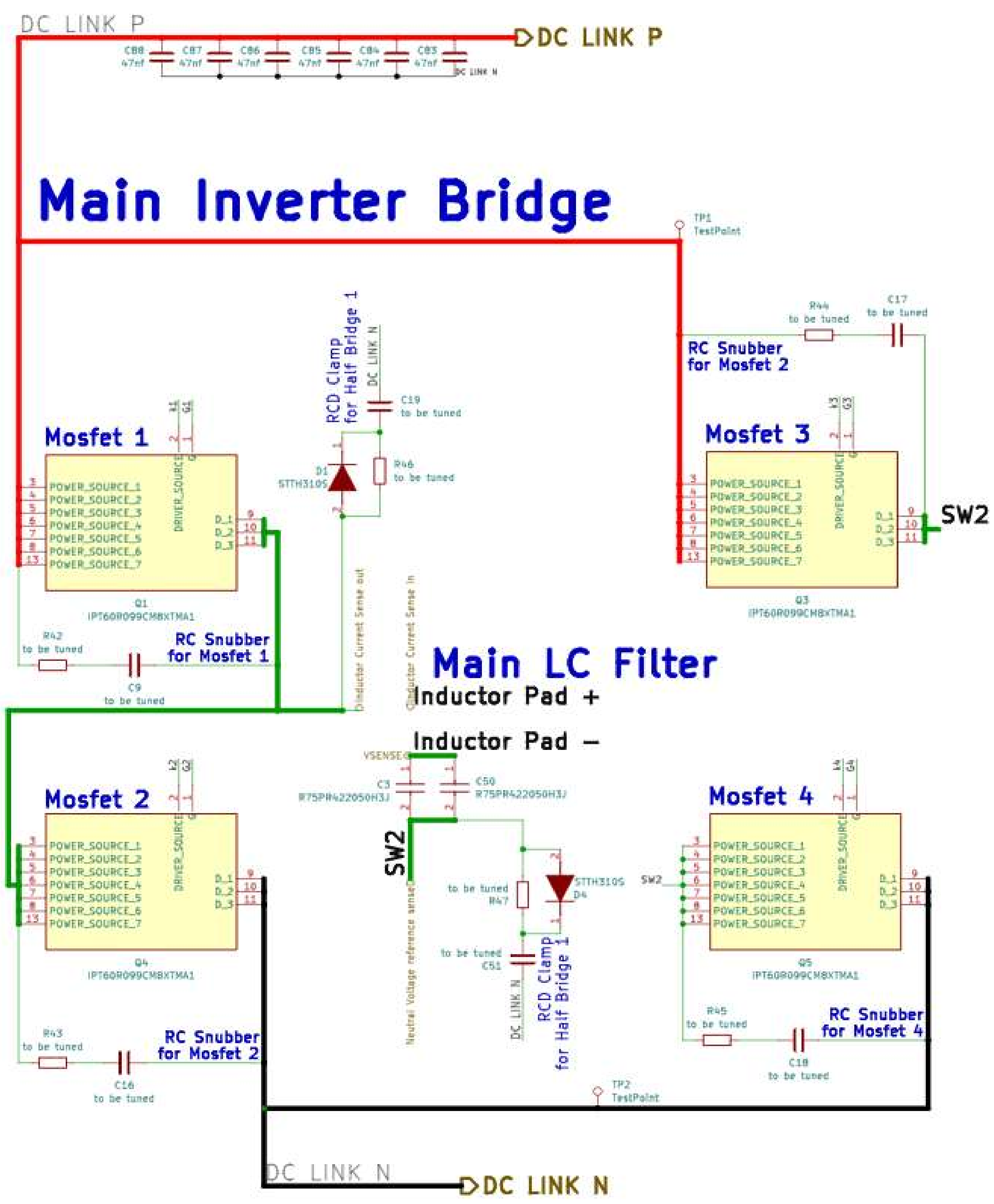

Fig .2 Inverter Schematic

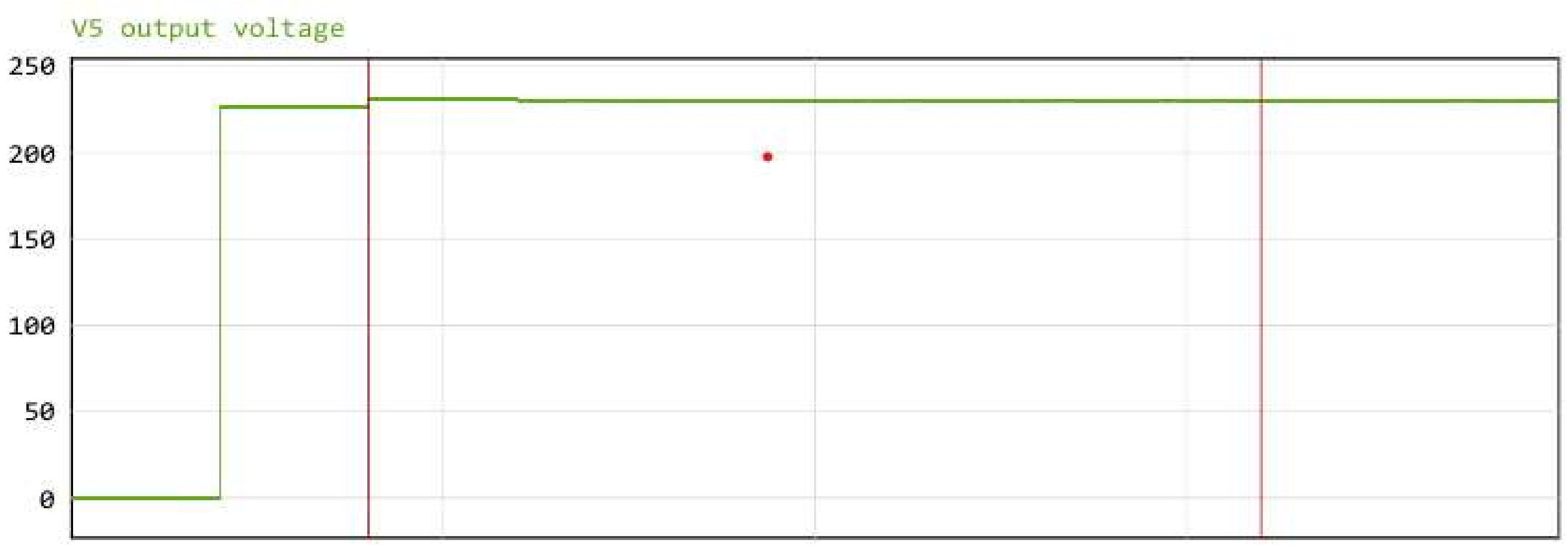

Fig .3 voltage output

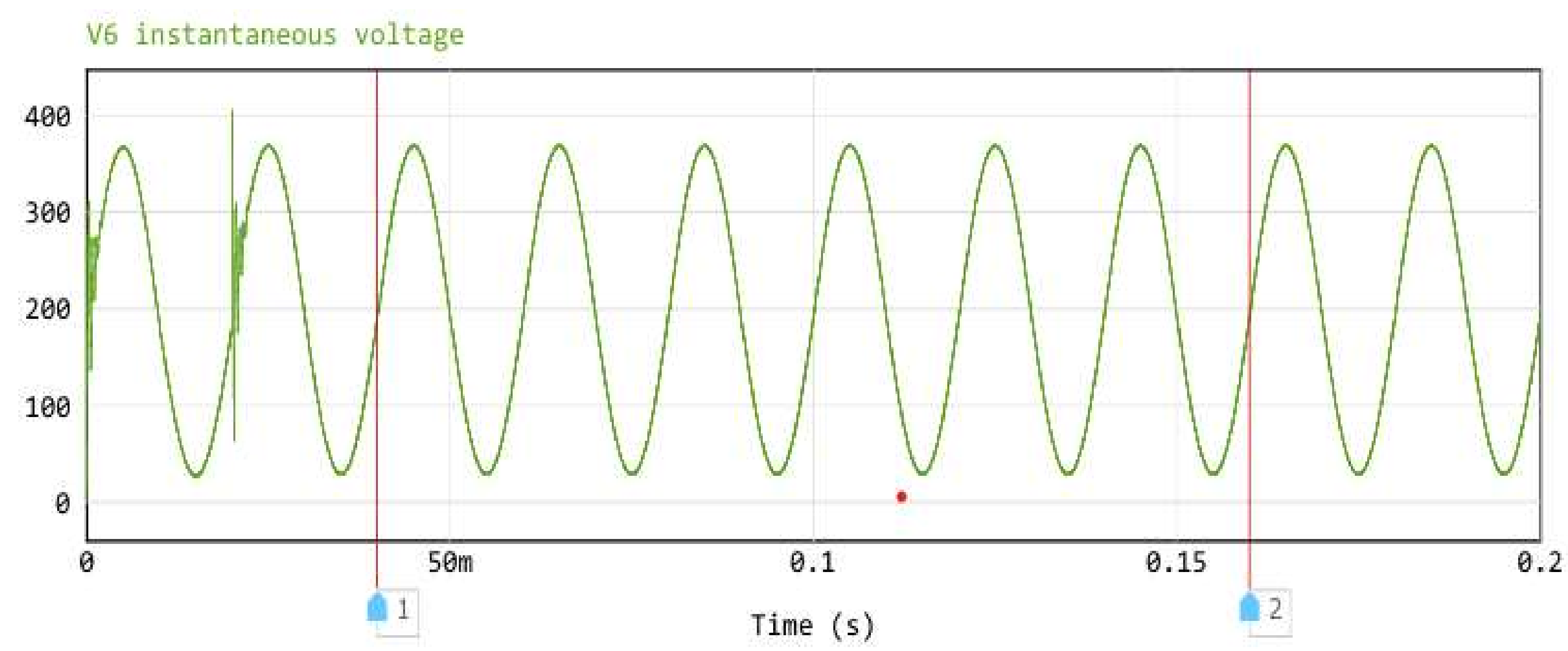

Fig .4 instantaneous voltage

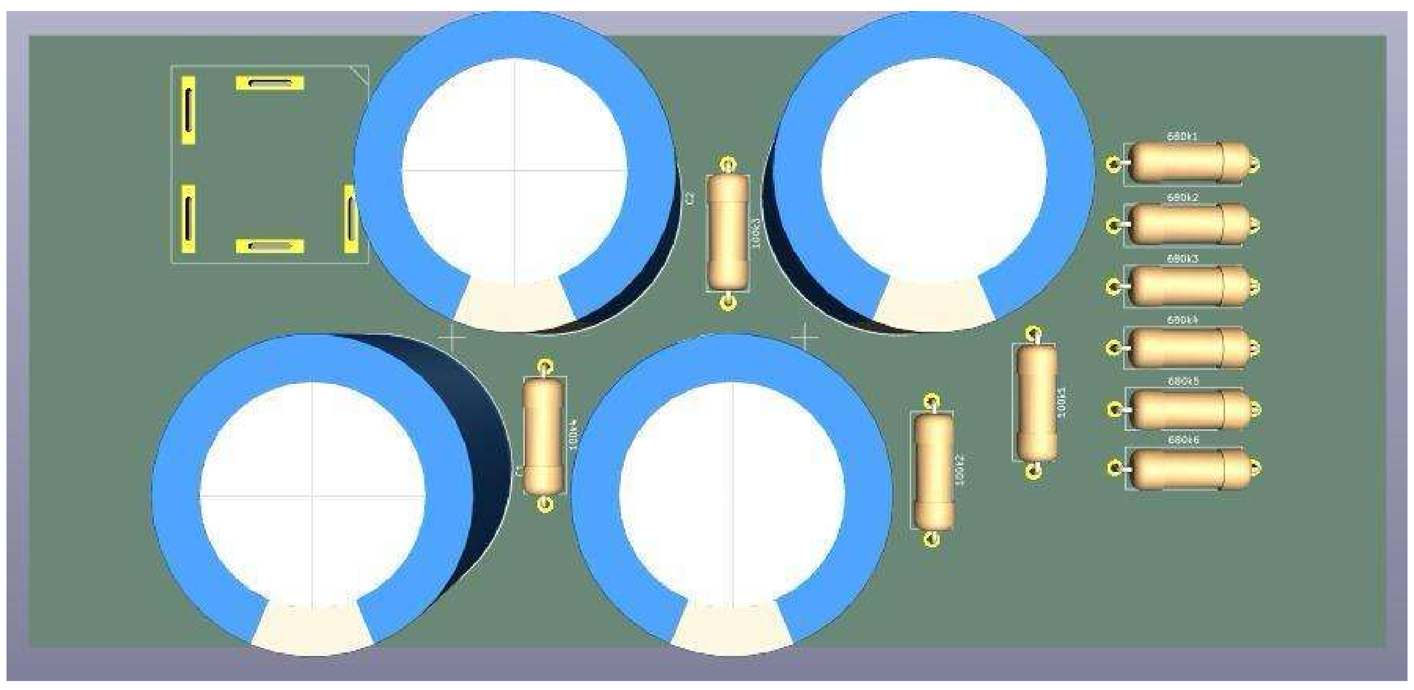

Fig .5 PCB design

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CALCULATIONS

1. OUTPUT POWER CALCULATION

Formula

$$P = V_{rms} I_{rms}$$

Given:
Output Power = 1000 W
Output Voltage = 230V RMS

Current Calculation

$$I_{rms} = \frac{P}{V_{rms}} = \frac{1000}{230}$$
$$= 4.35\ A$$

Output Current ≈ 4.35

2. DC LINK VOLTAGE CALCULATION

For single-phase inverter:

$$V_{DC} = \sqrt{2} V_{AC}$$

Given:
AC Output = 230V RMS

Calculation

$$V_{DC} = \sqrt{2} \times 230 = 325\ V$$

But your system uses:
400V DC Link
for better regulation and SPWM operation.

3. SWITCHING FREQUENCY

Given:
Switching Frequency = 20 kHz
Output Frequency = 50 Hz

Frequency Ratio

$$m_f = \frac{f_s}{f_o} = \frac{20000}{50} = 400$$

Result:
Frequency Modulation Ratio = 400
This improves waveform quality and reduces harmonics.

4. MODULATION INDEX

Formula

$$m_a = \frac{V_m}{V_c}$$

Given:
Modulation Index = 0.85

5. EFFICIENCY CALCULATION

Formula

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

Example:
If:
- Output Power = 1000W
- Input Power = 1100W

Calculation

$$\eta = \frac{1000}{1100} \times 100 = 90.9\%$$

Result:
Efficiency ≈ 91%

6. THD (TOTAL HARMONIC DISTORTION)

Formula

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots}}{V_1} \times 100$$

Mention:
THD < 3%
This indicates:
Better power quality
Lower harmonics
Cleaner output waveform

7. OUTPUT FREQUENCY

Formula

$$T = \frac{1}{f}$$

Given:
Frequency = 50 Hz

Calculation

$$T = \frac{1}{50} = 0.02\ s = 20\ ms$$

Result: Time Period = 20 ms

8. LC FILTER RESONANT FREQUENCY

VERY IMPORTANT calculation.
Formula

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Given:
L = 1.5 mH
C = 2.2 µF × 2 parallel
Total C = 4.4 µF

Calculation

$$f_r = \frac{1}{2\pi\sqrt{(1.5 \times 10^{-3})(4.4 \times 10^{-6})}} \approx 1961\ Hz$$

Result:
Resonant Frequency ≈ 1.96 kHz
This effectively removes 20kHz switching harmonics.

9. SHUNT RESISTOR VOLTAGE DROP

Given:

Current = 10A
Shunt Resistance = 5mΩ

Calculation

$$V = 10 \times 0.005 = 0.05\ V = 50\ mV$$

Result: Voltage Drop = 50 mV

10. POWER DISSIPATION IN SHUNT

Formula

$$P = I^2 R$$

Calculation

$$P = (10)^2 \times 0.005 = 0.5\ W$$

Result: Power Loss = 0.5W